



# Computational Concepts and their Assessment in Preschool Students: An Empirical Study

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## Abstract

Computational thinking (CT) is a multidimensional term that encompasses a wide variety of problem-solving skills related to the field of computer science. Unfortunately, standardized, valid, and reliable methods to assess CT skills in preschool children are lacking, compromising the reliability of the results reported in CT interventions. To surpass this limitation, we validated in a sample of 700 preschool students (5–6 years old) the Beginners Computational Thinking test Short-Form (BCTt-SF), an unplugged 12-item instrument that measures three of the most common computational concepts assessed in preschool research: sequences, loops, and conditionals. The theoretical model underpinning the BCTt-SF was supported by dimensionality assessment, which suggested that preschool students can be distinguished in terms of four specific abilities (i.e., sequences, simple loops, nested loops, and conditionals) and that all of these abilities were related by a general factor. We modeled this hierarchical structure with a bi-factor model that presented excellent psychometric properties, from good statistical fit indices to adequate reliability of the general ability. To take full advantage of this model, we created an online application in the Shiny platform (<https://computationalthinkingtests.shinyapps.io/SF-BCTt/>) for the seamless scoring of examinees by any teacher or researcher who uses the BCTt-SF to assess CT skills in preschool children. Finally, we demonstrated how the BCTt-SF can be used to test the impact of educational interventions for improving CT skills in preschoolers.

**Keywords** Computational thinking · Early years education · Evaluation methodologies · Psychometric validation · 21st century abilities

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## Introduction

Many instances of problem solving in daily life require mastering abilities closely related to terms used in computer science. For example, pattern recognition is an important skill for driving a vehicle to identify obstacles in traffic flow and adjust the route accordingly, whereas interpreting conditional rules (e.g., stopping *if* encountering a stop sign or at a yield sign *while* other vehicles keep approaching from either direction) are also essential aspects for driving that are reminiscent concepts from computer science.

However, the application of computational thinking (CT) skills is not restricted to highly standardized activities. Contrary, it comprises “the thought processes involved in formulating problems so their solutions can be represented as computational steps and algorithms” (Aho, 2012). As such, these processes can be transferred to nonprogramming domains and more nuanced situations (Weintrop et al., 2021). For instance, a person who is planning a vacation must break

the problem down into smaller, manageable parts (e.g., destination, transportation, accommodations, activities, budget, timing), which is an ability termed *decomposition* (Shute et al., 2017).

The fact that everyone uses CT abilities on a routinely basis and with different degrees of success has motivated many educators to suggest that CT should be a compulsory subject as worthy as reading, writing, and arithmetic (Wing, 2006) and that it should be assessed at all levels of education (El-Hamamsy et al., 2022b; Relkin & Bers, 2021; Stamatios, 2021), including preschool grades (Georgiou & Angeli, 2019; Stamatios, 2022). As a consequence, there has been a significant progress over the last years to promote CT in compulsory education (Kong, 2016; Manches & Plowman, 2015; Saxena et al., 2020; Wang et al., 2020; Weintrop et al., 2021), starting as early as primary school in some countries, such as New Zealand, South Korea, Canada, or Singapore, and in preschool, such as in Australia and Japan (Bocconi et al., 2016).

With the advent of this new academic curricula in which students acquire CT skills, instruments for monitoring their learning progress have been proposed recently. The assessment of CT skills is a highly valuable endeavor because it helps to assess strengths and gaps in the curricula, to identify children who need further support, and to provide feedback to teachers on the effectiveness of their teaching methods. Additionally, assessing CT skills can also serve to identify areas where students are excelling, prompting opportunities for enrichment and challenge for these students (Román-González et al., 2018). Alternatively, it may promote social equity by helping children with disabilities or special needs to prepare them for higher levels of education (Taylor, 2018).

Up to date, instruments for measuring CT skills span a wide audience, ranging from preschoolers to secondary school students (El-Hamamsy et al., 2022b; Relkin & Bers, 2021; Relkin et al., 2023; Román-González et al., 2018; Zapata-Cáceres et al., 2020, 2021). Notwithstanding, the development of rigorous testing procedures for the former group remains scarcer. This is a problem because an instrument designed for a specific age range is likely to be too difficult for those immediately younger, differing also in its capacity for disentangling students with different levels of skill. And more worrisome, a recent review of assessment methods in CT research for preschool children found that most of these methods were not standardized but created ad-hoc and did not undergo a proper validation and reliability analysis (Tang et al., 2020).

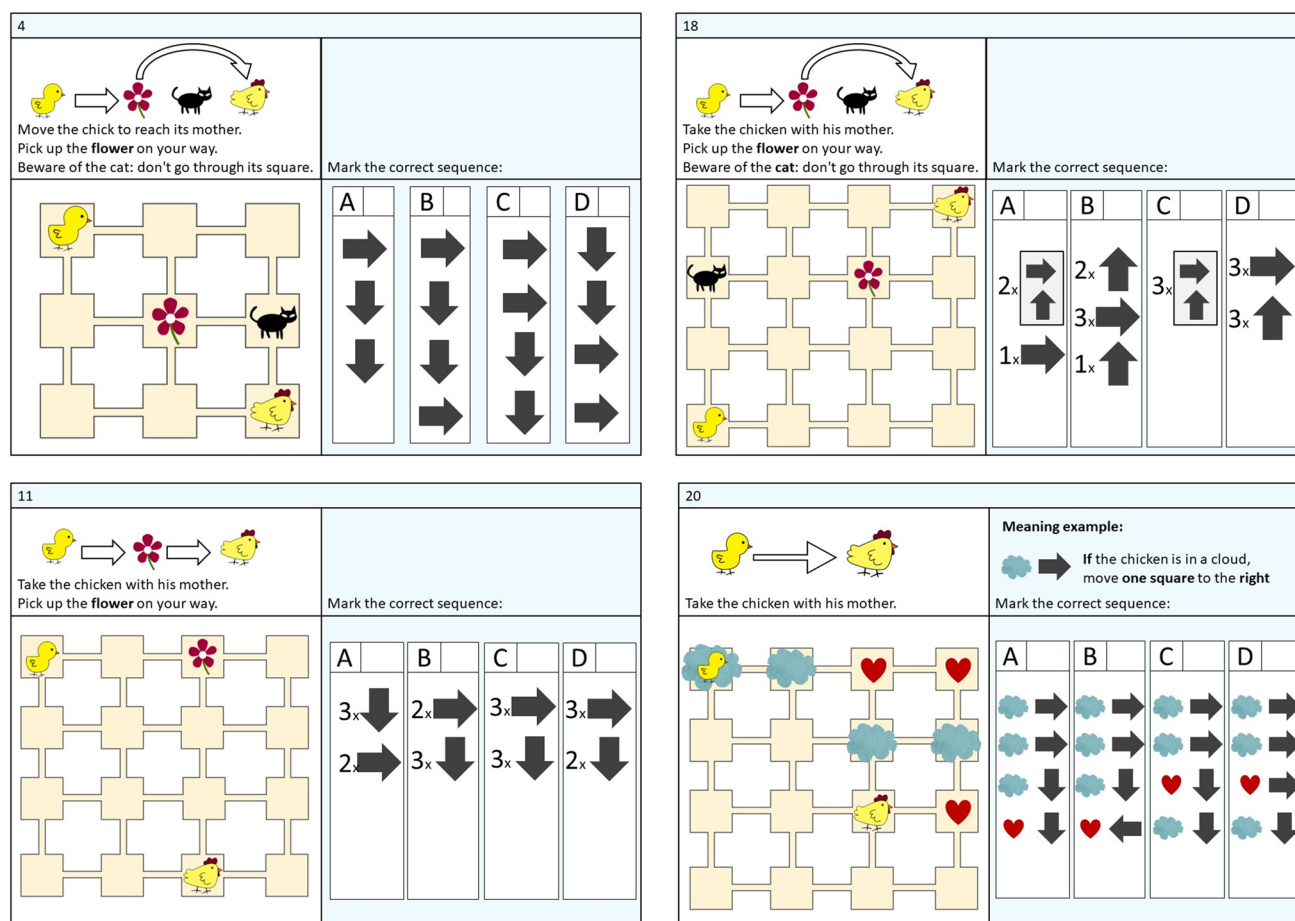
To overcome these limitations, we made a psychometric adaptation for preschool students of the Beginners Computational Thinking test (BCTt; Zapata-Cáceres et al., 2020), an instrument originally designed for children aged 5 to 9 years old that has been validated for measuring the understanding of three computational concepts in lower-primary

school: sequences, loops, and conditionals. Sequences are sets of instructions or steps that need to be taken in turn; loops are rules to run the same sequence a given number of times, either by repeating each time a sequential term (i.e., simple loop) or by repeating a number of times a block of sequential terms (i.e., nested loop). Finally, conditionals are indications to perform an action when either a specific circumstance is satisfied (i.e., if-then), when a condition is satisfied and another action when it is not (i.e., if-then-else), or as long as a condition remains satisfied (i.e., while).

In the BCTt, these three broad computational concepts are put into test in 25 items, categorized as either grid format (21 items) or canvas format (4 items). In the grid format items, the examinees are instructed to move an object (i.e., a baby chicken) across a grid to reach a target (i.e., the mother of the baby chicken) by spotting the correct sequential pattern from a list of four alternatives. Figure 1 provides an illustrative example of a typical grid item format in the BCTt, encompassing each of the aforementioned dimensions. Conversely, in canvas format items, the examinees are presented with a simple path drawing over a canvas and must select the correct sequential pattern that corresponds to the given path from another list of four alternatives. An important note is that, although the BCTt is mainly focused on the assessment of these computational concepts (e.g., sequences, loops, and conditionals), it is not limited to them. Contrary, computational practices such as testing and debugging are also required to solve the questions of the BCTt because the examinees need to inspect the four response alternatives (i.e., A, B, C, and D) to spot the erroneous ones.

The BCTt bears the potential to be applied in preschool settings because it was found to be most suited for low-percentile students in primary school (Zapata-Cáceres et al., 2021). However, the average time of completion in lower-primary school students was about 40 min (Zapata-Cáceres et al., 2021). Therefore, reducing the test length is necessary for preschool children to complete it, as they need a longer time to answer most of the items and are more prone to fatigue. Such, the aim of this study is three-fold:

- (1) First, to identify the nonoverlapping computational concepts that the students can differentially understand and upon which can be differentiated.
- (2) Second, to select the subset of items from the BCTt that consistently measures the target computational concepts in preschool students, discarding those that are tangential or measure spurious features. We also intend to shorten further the test by selecting the items that best encompass all the computational dimensions while retaining adequate psychometric properties of reliability and content validity, creating the Short-Form of the Beginners Computational Thinking test (BCTt-SF).



**Fig. 1** Depictions of four items belonging to the BCTt, each one taken from a different computational concept block. Upper-left: Sequence block; bottom-left: Simple Loop block; upper-right: Nested Loop block; bottom-right: Conditional block

- (3) Third, to create an online application where teachers can upload the responses of their students and automatically obtain the corresponding scores on all the computational dimensions assessed by the BCTt-SF.

### Computational Thinking in Preschoolers

Although no consensus definition of CT currently exists, there is agreement in that CT is a multidimensional term that encompasses a wide variety of problem-solving skills, including logical thinking, development of algorithms, data representation and manipulation, computer programming, and more (Shute et al., 2017). In other words, CT is not just about solving problems in a computational environment, but applying thought processes that lead to problem evaluation, simplification, and elaboration of strategies to find a solution using a set of skills that are related to computational science. However, the existence of multiple definitions, categorizations, and assessment proposals of CT abilities makes difficult for educators to decide what specific competences should be addressed in a given age group and how to do it.

For instance, while some authors warn about the scarcity of research with regards to the specific skills that preschoolers candominate (Relkin & Bers, 2021), others remark the lack of consensus about what computational concepts and practices should be taught to children at these early ages (McCormick & Hall, 2022).

According to Piaget's theory of cognitive development, preschool children fall under the preoperational stage of development that spans between 2 and 7 years old. This is a period when children acquire language and symbolic thinking. They learn to represent objects and events mentally without their physical presence and through images and words, prompting the appearance of abilities such as sequencing, pattern recognition, and algorithm design (Saxena et al., 2020).

In this vein, there exists some research in preschool children in relation to the specific concepts of sequences, loops, and conditionals that the BCTt measures. Specifically, there is consensus in that preschoolers can solve with accuracy simple loops problems (Angeli & Valanides, 2020; Bers et al., 2014; Saxena et al., 2020), although they struggle more than with complex sequences (Elkin et al., 2016). In any case, it is considered that this ability is already mastered in early

primary school (Relkin & Bers, 2021). With regards to their performance in solving problems with loops, research also shows that they can understand and apply with success this concept (Angeli & Valanides, 2020; Bers et al., 2014; Relkin & Bers, 2021). Lastly, there is also evidence that preschoolers can solve certain tasks involving conditional statements (Andrews et al., 2012; Diamond et al., 2005). In fact, it seems that they can do it much before than what it was previously thought (Ling et al., 2021), indicating that it is a concept that preschoolers can understand at an age of 3 years old.

### Test Development in Computational Thinking Research for Preschoolers

A systematic review conducted by Bakala et al. (2021) confirmed that sequences, loops, and conditionals were, in fact, the computational concepts most commonly assessed in the literature for the purpose of testing educational interventions. However, despite the recent development of many appealing educational programs and curricula for promoting the learning of CT skills in preschool children, the lack of valid instruments for testing them remains as one of the biggest problems in the field (Bakala et al., 2021; Tang et al., 2020). Moreover, in order to make an objective comparison between the many different educational interventions that have been developed in the field, it is necessary to operationalize the computational concepts and practices that are relevant for different age groups and create standardized instruments accordingly (McCormick & Hall, 2022). Thus, we currently do not know to what extent the rubrics, tasks, and tests that conform these assessment methods measure the intended computational constructs and, ultimately, if they are valid for the purpose of scoring students. As a consequence of this lack of validated instruments, the fair comparison of different CT interventions is neither possible. Moreover, the samples recruited in these studies were generally small, precluding even the application of reliable psychometric analyses.

Another problem is that most of the reviewed instruments were plugged (i.e., they involved tasks where students needed to interact with electronic devices, write or interpret code, program robots, and perform other activities that required specific computational knowledge). Unfortunately, these kinds of plugged assessments have drawbacks. When using technological environments to measure CT skills, children who lack the necessary coding knowledge to participate cannot be assessed, prompting the risk of conflating CT abilities with programming skills. Moreover, some children may outperform others because of more familiarity or prior exposure to technological environments and a more developed fine-motor functioning. An additional disadvantage of plugged methods for assessment is that they are costly

because they require materials (e.g., physical kits) and the individual observation of each child. Furthermore, plugged tasks are often assessed with portfolios and rubrics, which can also be problematic due to the subjectivity of the teacher who scores the behavior of the children.

For these reasons, many of the new psychometric instruments that are being developed take an unplugged design based on traditional tests (El-Hamamsy et al., 2022b; Li et al., 2021; Relkin & Bers, 2021; Relkin et al., 2020, 2023; Román-González et al., 2015; Zapata-Cáceres et al., 2021; Zhang & Wong, 2023). Consider the TechCheck (Relkin et al., 2020), an unplugged 15-item test designed for preschool students that measures six different computational concepts and practices: algorithms, modularity, debugging, hardware/software, control structures, and representation. The TechCheck does not require any coding knowledge and can be easily scored. In addition, it was adapted to the population of preschool students using a sample of 395 students, yielding the revised version TechCheck-K (Relkin & Bers, 2021; Relkin et al., 2023).

Another example is the BCTt (Zapata-Cáceres et al., 2021). As pointed out by Bakala et al. (2021), instruments like the BCTt are advantageous because many of the activities that were assessed in the literature were closely related to the tasks used during the educational program, whereas the BCTt offers measures that are independent from the devices used during the intervention, providing information about the transferable properties of CT skills. Moreover, the BCTt is an instrument created from adaptations of previous tests widely used in higher education levels that have undergone validity and reliability assessments (Table 1). In fact, we are not the first authors who use the BCTt to create an adaptation for students of a different age group. The BCTt has already been successfully adapted for upper-primary school students by increasing item difficulty, giving raise to the competent Computational Thinking test (cCTt; El-Hamamsy et al., 2022a). As such, the theory underpinning the BCTt is well consolidated, and the BCTt itself serves as a benchmark for other tests as well as a template for the development of new tests (El-Hamamsy et al., 2022a).

To determine the internal validity of the BCTt and to accomplish a reliable adaptation of the BCTt with a suitable test length for preschool students, we pursued the three objectives outlined above: First, we determined which computational concepts can be distinctly assessed in a sample of preschool students. Second, we undertook a purification process in which screening out the items that do not measure the intended computational concepts. As a result, we selected the most informative items while maintaining adequate validity and reliability properties, resulting in a shortened version of the test (BCTt-SF). Third, we provided an online applica-

**Table 1** Review of Computational Thinking tests directly related to the BCTt-SF in chronological order

| Test                                                                 | Format                                          | Target group                         | CT concepts                                                                                                                                               | Psychometric assessment                                                                                                                                                                        | Sample                                                                                                                                                   |
|----------------------------------------------------------------------|-------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| CTt (Román-González et al., 2015, 2017, 2018; Guggemos et al., 2022) | 28 items                                        | Secondary school (10–16 years old)   | Sequences, Finite Loops, Loops over a condition, Conditionals (i.e., If-then, If-else, and While statements), and Functions                               | Expert validation, Reliability estimation within CTT (i.e., Cronbach alpha) and Item Response Theory, Criterion validity, Predictive validity                                                  | 400 students (Román-González et al., 2015) 1251 students (Román-González et al., 2017) 314 students (Román-González et al., 2018; Guggemos et al., 2022) |
| BCTt (Zapata-Cáceres et al., 2020)                                   | 25 items                                        | lower-Primary school (5–7 years old) | Sequences, Simple Loops, Nested Loops, and Conditionals (i.e., If-then, If-else, and While statements)                                                    | Reliability estimation within CTT (i.e., Cronbach alpha)                                                                                                                                       | 299 students                                                                                                                                             |
| cCTt (El-Hamamsy et al., 2022a)                                      | Three versions of 25, 17, and 15 items          | upper-Primary school (7–9 years old) | Sequences, Simple Loops, Nested Loops, Conditionals (i.e., If-then, If-else, and While statements), Complex (i.e., combinations of the previous concepts) | Expert Validation, Model fit assessment with CFA, Item calibration and reliability estimation within IRT (i.e., item information), Reliability estimation within CTT (i.e., Cronbach alpha)    | 1519 students                                                                                                                                            |
| BCTt-SF (current study)                                              | Selection of 20 items Final version of 12 items | Preschool students (4–6 years old)   | Sequences, Simple Loops, Nested Loops, Conditionals (i.e., If-then and If-then-else)                                                                      | Dimensionality assessment, Model fit assessment with CFA, Item calibration and reliability estimation with MIRT (i.e., item information), and automatic test scoring via an online application | 700 students                                                                                                                                             |

*Note.* CTt: Computational Thinking test; BCTt: Beginners Computational Thinking test; cCTt: competent Computational Thinking test; BCTt-SF: Short-Form of the Beginners Computational Thinking test; CTT: Classical Test Theory



tion for the automatic and seamless scoring of any preschool examinee.

## Methods

In this section, we describe the details of the data collection and the psychometric analyses that we carried out to develop the adaptation of the BCTt for preschool students.

### Participants and Data Collection

A sample of 700 preschool students aged 5 to 6 years old recruited from 8 classrooms across several Spanish regions participated in the current study. The sample was divided in two groups: quasi-experimental and control. The quasi-experimental group encompassed 595 children whose teachers subscribed voluntarily to an educational program promoted by the Spanish Ministry of Education and Vocational Training.<sup>1</sup> The aim of this project was investigating the impact of the unplugged CT activities in the development of the CT skills of children. The educational intervention, termed *School of Computational Thinking and Artificial Intelligence*, included a training phase for the teachers (i.e., from 13 November 2020 to 1 February 2021) and a curriculum implementation phase for the children (i.e., from 1 March to 31 May 2021). The training phase was composed by four blocks. In the first block, CT was introduced to the teachers, who learned the definitions of several concepts such as sequences, algorithms, logical thinking, abstraction, decomposition, parallelism; in the second block, they completed different unplugged activities involving these concepts to acquire familiarity with them. In the third block, they learned how to analyze the didactic value of the unplugged activities using *Cody & Roby*,<sup>2</sup> a game in which Cody (i.e., the programmer) creates instructions that Roby (i.e., the robot) must execute. The final block was dedicated to the presentation of unplugged activities for the teaching of Artificial Intelligence (Touretzky et al., 2019). In total, the teachers dedicated 30 h to complete all of these training blocks.

Once the training phase was completed, the teachers implemented their custom educative intervention in at least five lectures during the following months. This intervention was integrated into standard subjects of the curriculum (i.e., mathematics, linguistics, natural and social sciences, physical education, foreign language, and music) following the guide proposed by AI4K12<sup>3</sup> (Nishida et al., 2009). In this manner, the CT activities also contributed to the learning of

specific content from these subjects. During the implementation phase, the children participated in five to ten CT teaching sessions that included unplugged CT activities. Only the children whose teachers completed both phases of the program were considered in the study.

On the other hand, the control group was composed by 105 students recruited with the help of the Education Departments of different Spanish regions. The teachers in the control group also participated voluntarily in the study, but they did not undergo the educational intervention. The only requirement for the students in the control group to participate in the study was the absence of unplugged CT activities in their daily teaching during the current and previous academic courses.

After completing the implementation phase in the participating schools, all students, both in the quasi-experimental group and the control group, took the BCTt to measure their CT skills. The BCTt is an instrument that has been widely used in primary school students (Zapata-Cáceres et al., 2020; Zapata-Cáceres & Fanchamps, 2021) and has been found to display good reliability in terms of internal consistency, with a Cronbach's alpha coefficient of .824 (Zapata-Cáceres et al., 2020). As such, the BCTt may be a reliable measure to assess the CT skills of the students in the current study.

Regarding the test administration, the BCTt was taken during a class period, with students having a total of 45 min to answer the questions, in addition to a few minutes of prior explanation about the test scoring. The BCTt is a paper-based test, so each student had an individual printed copy of the test. To ensure that the instructions given to the students before taking the test were as similar as possible in all the schools, a video was recorded<sup>4</sup> explaining each type of question of the test. This video was made available to teachers so they could show it to their students in class just before they took the test. The teachers were also provided with an action protocol and a document with answers to frequently asked questions.

In the end, the educational intervention used in our study was not designed to teach the children any particular concept included in the BCTt. Consequently, we expect that the results from the BCTt indicate their overall CT learning, rather than being affected by specific knowledge acquired from test preparation.

### Data Analysis and Psychometric Validation

We first inspected the data with descriptive statistics (Table 2). These included the proportion in which every option to each item was selected and the cumulative missingness at each item. Table 2 also displays the qualitative characteristics of the items composing the BCTt, which includes their format (i.e., grid vs. canvas) and the particular mistakes that produce the selection of erroneous response alternatives (i.e.,

<sup>1</sup> <https://intef.es/>

<sup>2</sup> <http://www.codeweek.it/cody-roby-en/>

<sup>3</sup> <https://ai4k12.org/>

<sup>4</sup> <https://vimeo.com/544462776/34e6be254a>

**Table 2** Item features and error profiles for each response alternative. The proportion in which each alternative was chosen is also presented within parentheses

| Block       | Q  | Format       | Response alternatives |           |           |           | Missingness |     | Error Profiles |                 |                    |                             |                          |                 |             |
|-------------|----|--------------|-----------------------|-----------|-----------|-----------|-------------|-----|----------------|-----------------|--------------------|-----------------------------|--------------------------|-----------------|-------------|
|             |    |              | A                     | B         | C         | D         | Prop        | WD  | Diff           | 1               | 2                  | 3                           | 4                        | 5               | 6           |
| Sequences   | 1  | Grid (3 × 3) | ✓ (.89)               | 1 (.05)   | 6 (.03)   | 1 (.03)   | .00         | .00 | —              | Wrong direction | Insufficient steps | Fails to avoid the obstacle | Fails to pick the object | —               | Other error |
|             | 2  | Canvas       | 1 (.05)               | 1 (.05)   | 1 (.05)   | ✓ (.81)   | .04         | .00 | —              |                 |                    |                             |                          |                 |             |
|             | 3  | Grid (3 × 3) | 3 (.05)               | 2/3 (.04) | ✓ (.82)   | 1 (.05)   | .03         | .00 | —              |                 |                    |                             |                          |                 |             |
|             | 4  | Grid (3 × 3) | 2 (.06)               | ✓ (.77)   | 3/4 (.05) | 4 (.09)   | .04         | .00 | -0.58          |                 |                    |                             |                          |                 |             |
|             | 5  | Grid (4 × 4) | 3/4 (.07)             | ✓ (.77)   | 3 (.07)   | 2/3 (.07) | .02         | .00 | -0.10          |                 |                    |                             |                          |                 |             |
|             | 6  | Canvas       | 1 (.06)               | 1 (.06)   | 2 (.09)   | ✓ (.72)   | .07         | .01 | 1.06           |                 |                    |                             |                          |                 |             |
| Simple Loop | 7  | Grid (3 × 3) | 2 (.12)               | ✓ (.76)   | 5 (.06)   | 5 (.02)   | .04         | .01 | 3.89           | Wrong direction | Insufficient steps | Fails to avoid the obstacle | Fails to pick the object | Leaves the grid | Other error |
|             | 8  | Grid (3 × 3) | ✓ (.74)               | 5 (.10)   | 5 (.08)   | 1 (.04)   | .05         | .01 | 0.89           |                 |                    |                             |                          |                 |             |
|             | 9  | Grid (3 × 3) | 3 (.11)               | 3 (.06)   | 6 (.05)   | ✓ (.73)   | .05         | .02 | 2.19           |                 |                    |                             |                          |                 |             |
|             | 10 | Grid (4 × 4) | 2 (.09)               | 2 (.05)   | ✓ (.75)   | 2 (.04)   | .07         | .03 | 0.12           |                 |                    |                             |                          |                 |             |
|             | 11 | Grid (4 × 4) | 4 (.18)               | ✓ (.63)   | 6 (.10)   | 6 (.04)   | .06         | .03 | -              |                 |                    |                             |                          |                 |             |
|             | 12 | Grid (4 × 4) | ✓ (.61)               | 2 (.13)   | 2 (.05)   | 2 (.13)   | .08         | .03 | 1.56           | Wrong direction | Insufficient steps | Fails to avoid the obstacle | Fails to pick the object | Leaves the grid | Other error |
| Nested Loop | 13 | Canvas       | 1 (.09)               | 1 (.06)   | ✓ (.68)   | 1 (.09)   | .09         | .04 | 0.09           |                 |                    |                             |                          |                 |             |
|             | 14 | Grid (4 × 4) | 3 (.26)               | ✓ (.41)   | 3/5 (.05) | 2 (.17)   | .11         | .05 | 0.72           |                 |                    |                             |                          |                 |             |
|             | 15 | Grid (4 × 4) | 4 (.32)               | 2 (.12)   | ✓ (.23)   | 2 (.21)   | .13         | .07 | 7.81           |                 |                    |                             |                          |                 |             |
|             | 16 | Grid (4 × 4) | 4 (.22)               | 4/5 (.15) | 2 (.09)   | ✓ (.38)   | .17         | .10 | -0.24          |                 |                    |                             |                          |                 |             |
|             | 17 | Canvas       | ✓ (.69)               | 1 (.06)   | 1 (.04)   | 1 (.04)   | .16         | .11 | 1.59           |                 |                    |                             |                          |                 |             |
|             | 18 | Grid (4 × 4) | 2 (.23)               | 3 (.15)   | ✓ (.33)   | 4 (.11)   | .17         | .11 | 3.67           |                 |                    |                             |                          |                 |             |

Table 2 continued

| Block        | Q  | Format       | Response alternatives |         |         |         | Missingness |     |       | Error Profiles            |                 |                    |                                   |   |   |
|--------------|----|--------------|-----------------------|---------|---------|---------|-------------|-----|-------|---------------------------|-----------------|--------------------|-----------------------------------|---|---|
|              |    |              | A                     | B       | C       | D       | Prop        | WD  | Diff  | 1                         | 2               | 3                  | 4                                 | 5 | 6 |
| If-then      | 19 | Grid (3 × 3) | ✓ (.51)               | 1 (.11) | 2 (.08) | 5 (.11) | .19         | .13 | 2.35  | Follows a simple sequence | Wrong direction |                    | —                                 |   |   |
|              | 20 | Grid (4 × 4) | 1 (.27)               | 2 (.14) | ✓ (.27) | 2 (.25) | .25         | .15 | 1.75  |                           |                 |                    |                                   |   |   |
| If-then-else | 21 | Grid (4 × 4) | 2 (.15)               | 3 (.21) | 2 (.14) | ✓ (.25) | .25         | .18 | 0.30  | —                         | Wrong direction | Insufficient steps | —                                 | — |   |
|              | 22 | Grid (4 × 4) | 2 (.27)               | ✓ (.25) | 2 (.09) | 2 (.11) | .28         | .20 | -0.18 |                           |                 |                    |                                   |   |   |
| While        | 23 | Grid (3 × 3) | ✓ (.51)               | 2 (.11) | 2 (.08) | 2 (.06) | .24         | .22 | 1.80  | —                         | Wrong direction | —                  | Fails to move from starting point |   |   |
|              | 24 | Grid (4 × 4) | 4 (.30)               | ✓ (.26) | 2 (.09) | 2 (.07) | .28         | .25 | 1.89  |                           |                 |                    |                                   |   |   |
|              | 25 | Grid (4 × 4) | 2 (.16)               | 2 (.13) | ✓ (.31) | 2 (.10) | .31         | .31 | —     |                           |                 |                    |                                   |   |   |

*Note.* Q: Question; Prop: proportion of missing responses; WD: cumulative proportion of withdrawal from the item onward; Diff: Difference between the sum score of the examinees that answered all the questions to the sum score of those who withdrew after answering item; Wrong direction: Moves in an opposite direction or to a farther place with respect to the target; Insufficient steps: The number of steps are not enough to reach the target; Fails to avoid the obstacle: Passes through the black cat; Fails to pick the object: Does not pick the flower



error profiles). We identified six error profiles across all the sequences and loop blocks: Moving in an opposite direction or to a farther place with respect to the target (profile 1); taking an insufficient number of steps to reach the target (profile 2); failing to avoid an obstacle (profile 3); failing to pick a required object (profile 4); leaving the grid (profile 5); and committing another kind of error (profile 6). The error profiles identified in the items from the conditional block are more idiosyncratic and do not overlap with the previous ones. Although the error profiles will not be involved in any quantitative analysis, we included these profiles because we think they would help teachers to identify problematic patterns of reasoning in children, facilitating the design of teaching methods focused on mitigating these errors.

The second step consisted of identifying the dimensionality and structural configuration of the test in the sample. Dimensionality assessment is a particular complex task since model misspecification is an important obstacle towards understanding the underlying organization of the data (MacCallum, 2003). In other words, all statistical models ignore many of the nuances that explain some item variance (e.g., fatigue, distraction, pressure, acquiescence). As a result, empirical data is messy, difficulting its alignment with theory even when the main features of such a theory are true. Fortunately, Exploratory Graph Analysis (EGA; Golino & Epskamp, 2017b) is a recent dimensionality tool that can be used to uncover the underlying organization of the traits underpinning the responses to a set of items. It employs a regularization procedure designed to *clean* these nuances from the data. More specifically, EGA fits a Gaussian Graphical Model (GGM) with GLASSO regularization (Friedman et al., 2008; Golino & Epskamp, 2017b) to estimate sparse correlations between the items (i.e., their pair-wise linear association after taking out the shared variance with the remaining items) and then applies a clustering algorithm (e.g., Louvain) to classify each item in different, nonoverlapping groups. The result of this analysis is usually visualized as a graph, where each node is an item, the node color represents a cluster, and the thickness of the edges connecting them indicates the strength of their linear relationship after conditioning on all the remaining items.

Once this first dimensionality analysis was completed, a hierarchical version of EGA was performed (hierEGA; Jiménez et al., 2023). A score on each cluster was computed for each participant, and the GGM with Louvain clustering was applied again using these scores as input. As a result, the clusters computed in the first step (i.e., first-order clusters) were themselves clustered in higher-order clusters. This last step was used to identify the hierarchical organization of the data. In other words, we checked whether the first-order clusters were embedded within one or more higher-order clusters. Overall, we expected to find well-defined first-order clusters composed by items related to sequences, loops, and condi-

tionals, which are the three broad blocks of computational concepts used to design the items of the BCTt, as well as a single higher-order cluster (i.e., general factor) accounting for the relationships between them.

To obtain a more fine-grained analysis of the items and to develop a proper item scoring method, Multidimensional Item Response Theory (MIRT) was used. More concretely, we used a bayesian Three-Parameter Logistic (3PL) bi-factor model to estimate the probability of succeeding an item. According to the bayesian 3PL bi-factor model, the conditional probability of an examinee correctly scoring an item  $j$  is defined by the inverse logit function (Cai et al., 2011),

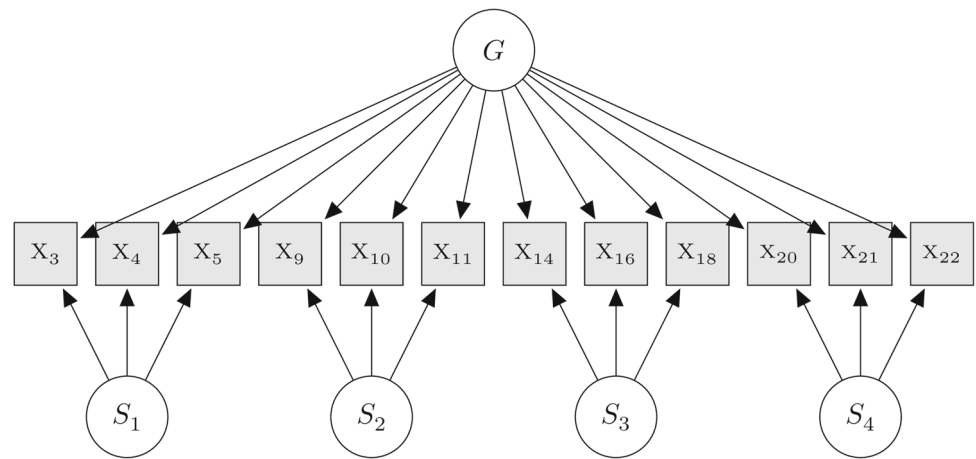
$$p_j = c_j + \frac{1 - c_j}{1 + \exp(-[\theta_g \alpha_{gj} + \theta_s \alpha_j + d_j] + G \times \eta_j)}, \quad (1)$$

where  $c_j$  is a parameter indicating the basal probability of succeeding item  $j$  by guessing,  $\theta_g$  and  $\theta_s$  are the values of the general and specific abilities with respective slope or *discrimination* parameters  $\alpha_{gj}$  and  $\alpha_j$ , and  $d_j$  is a location parameter for item  $j$ , usually termed the *easiness* parameter.  $G$  is an indicator variable that is 1 if the examinee belongs to the quasi-experimental group and 0 otherwise. This manner, the  $\eta_j$  indicated the difference in location between the item characteristic curves of the quasi-experimental and control groups, taking into account the effect of the educational intervention in the quasi-experimental group. Thus, these  $\eta_j$  parameters are akin to the ones that are commonly used to test for uniform differential item functioning.

To ascertain that the model represented accurately the observed data, we performed posterior predictive checks of the total score distribution for the quasi-experimental and control groups and for the probability of succeeding each item. Regarding the model parameter estimates, we flagged guessing estimates higher than .25, as they would imply that the basal probability of correctly answering the item is compatible with choosing a response alternative by chance.

Taking together the results of the analyses, we selected three items from each block of computational concepts, focusing on the items that were most informative for the general factor across a range of abilities that were not too high or too low. We also considered desirable that the items loaded on the specific dimensions and did not have high guessing probabilities. The selected items conformed the BCTt-SF, whose underlying bi-factor structure is anticipated in the diagram of Fig. 2. To assess the adequacy of the final 12-item selection, we repeated all the psychometric analyses on the items comprising the BCTt-SF. Finally, to take full advantage of the final statistical model, we created an application in the Shiny platform to automatically score examinees from their patterns of success/failure responses (<https://computationalthinkingtests.shinyapps.io/>

**Fig. 2** Diagram representation of the statistical model underlying the BCTt-SF.  $G$  = General Factor of Computational Concepts;  $S_1$  = Sequence factor;  $S_2$  = Simple Loop factor;  $S_3$  = Nested Loop factor;  $S_4$  = Conditional factor



**SF-BCTt**). This scoring method was based on Expected a Posteriori Estimation (EAP).

In the end, to estimate the effect of the educational intervention on the measured CT skills, we used the posterior samples of the ability estimates for the quasi-experimental and control groups. For each sample of the posterior distribution of the general ability, we computed the Cohen's  $d$  between the groups. As a result, we obtained the Expected a Posteriori value of the Cohen's  $d$  for the educational intervention and corresponding credible interval at the 95% level.

### Transparency and Openness

All the analyses were performed in the R programming language. For conducting the EGA analyses, we employed the EGAnet package (Golino & Christensen, 2022), version 1.1.1. We used the package lavaan (Rosseel, 2012), version 0.6-13, for fitting the CFA models described in the appendix and the package cmdstanr (Gabry & Češnovar, 2021), version, 0.4.0, to execute the bayesian bi-factor MIRT model. All

the materials, including the full data and scripts to reproduce the psychometric analyses, are available in the Open Science Framework repository at [https://osf.io/3xn7a/?view\\_only=7190e194469f44119aa9174b768a406d](https://osf.io/3xn7a/?view_only=7190e194469f44119aa9174b768a406d).

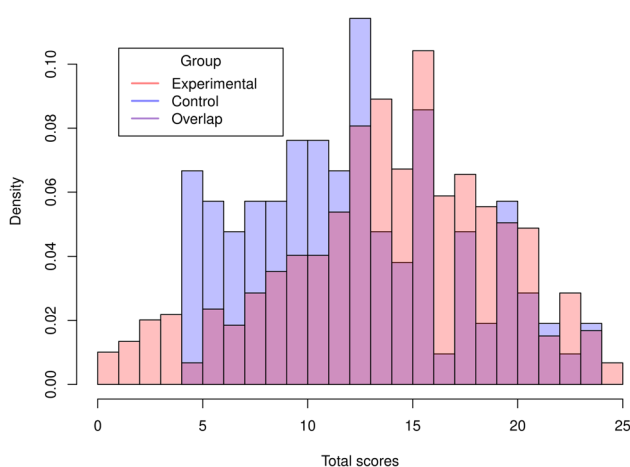
### Results

The results of the analyses carried out with the sample of 700 preschool students are presented next. We first present the results of the preliminary analyses involving descriptive statistics, followed by the results of the dimensionality assessment and bi-factor MIRT model. Finally, we review the psychometric properties of the shortened version of the BCTt.

#### Preliminary Analyses

The distributions of the total scores were bell-shaped (Fig. 3), slightly right-skewed for the quasi-experimental group and left-skewed for the control group. Neither floor nor ceiling effects were observed. The most common score in the quasi-experimental group was 16, whereas in the control group, it was 12. The quasi-experimental group also presented a higher median and interquartile range (i.e., median = 15; IQR = [11, 18]) than the control group (i.e., median = 12; IQR = [9, 16]). The mean scores for the quasi-experimental and control groups were closer, 14.32 and 12.66, respectively. A Welch two samples  $t$ -test showed that this difference was significant at the 95% confidence level ( $t = 3.09$ ;  $p$ -value = .0024).

The difficulty of the items was also progressive, with the highest proportion of success happening in the sequence block (.79), followed by the simple loop block (.72), the nested loop block (.48), and the block of conditionals (.34). The pattern of proportions of success was similar in the quasi-experimental and control groups, but the former usually



**Fig. 3** Histograms of total scores in the BCTt for the quasi-experimental and control groups

presented higher proportions of success (Fig. 4), especially for the items in the sequence and simple loop blocks.

We tested the significance of the difference between the proportions of correct responses between the quasi-experimental and control groups (*diff*). At a confidence level of 95%, we found significant differences in items 2 (*diff* = .17;  $\chi^2 = 16.32$ ;  $p = .000$ ), 5 (*diff* = .15;  $\chi^2 = 10.393$ ;  $p = .001$ ), 6 (*diff* = .20;  $\chi^2 = 16.25$ ;  $p = .000$ ), 7 (*diff* = .10;  $\chi^2 = 4.40$ ;  $p = .036$ ), 9 (*diff* = .12;  $\chi^2 = 5.86$ ;  $p = .016$ ), 10 (*diff* = .11;  $\chi^2 = 4.75$ ;  $p = .029$ ), 14 (*diff* = .16;  $\chi^2 = 8.87$ ;  $p = .003$ ), and 16 (*diff* = .11;  $\chi^2 = 4.20$ ;  $p = .040$ ), all of them in favor of the quasi-experimental group. Interestingly, no item from the conditional block presented a significant difference in the proportions of correct response between the groups.

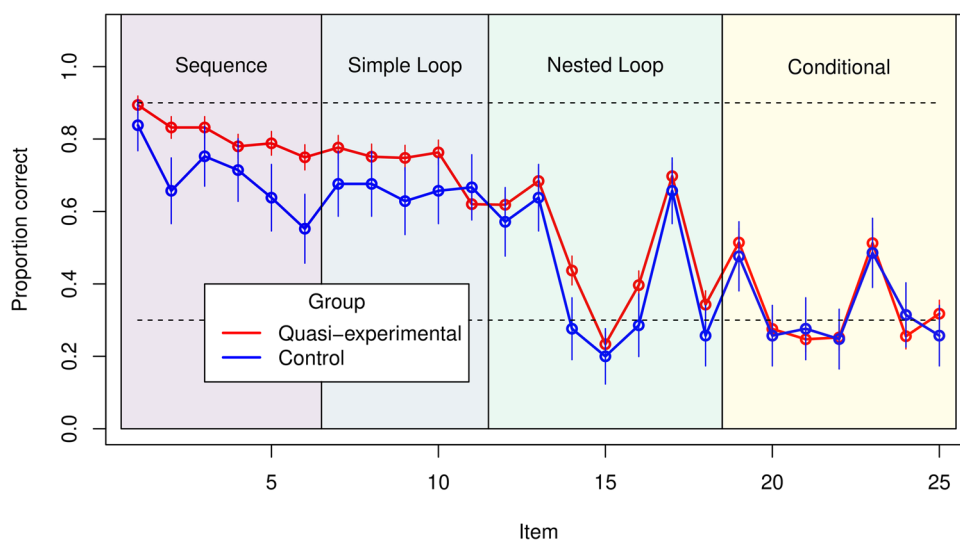
Four children attained the maximum possible score (i.e., 25), and four obtained the minimum score (i.e., 0). From the latter group, two examinees provided only one (incorrect) answer to the first item and did not respond to the others. This prompted us to take a closer look at missing data. The proportion of missing responses (Prop) was similar between the control and quasi-experimental groups. The overall Prop increased progressively from item 1 (Prop = 0) to item 25 (Prop = .31) with no steep increase from one item to the next (Table 2). This observation was compatible with examinees progressively withdrawing the test, so we also calculated the cumulative proportion of withdrawal (WD; Table 2) when faced each item. Indeed, this was the main reason behind missing responses. After item 14, most of the missing responses in each of the subsequent items came from examinees that had withdrawn the test. This turning point could be explained by the fact that item 15 was significantly more difficult than all the previous ones, possibly causing despair in examinees with lower scores. Thereby, to check whether test withdrawal was related to item difficulty, we

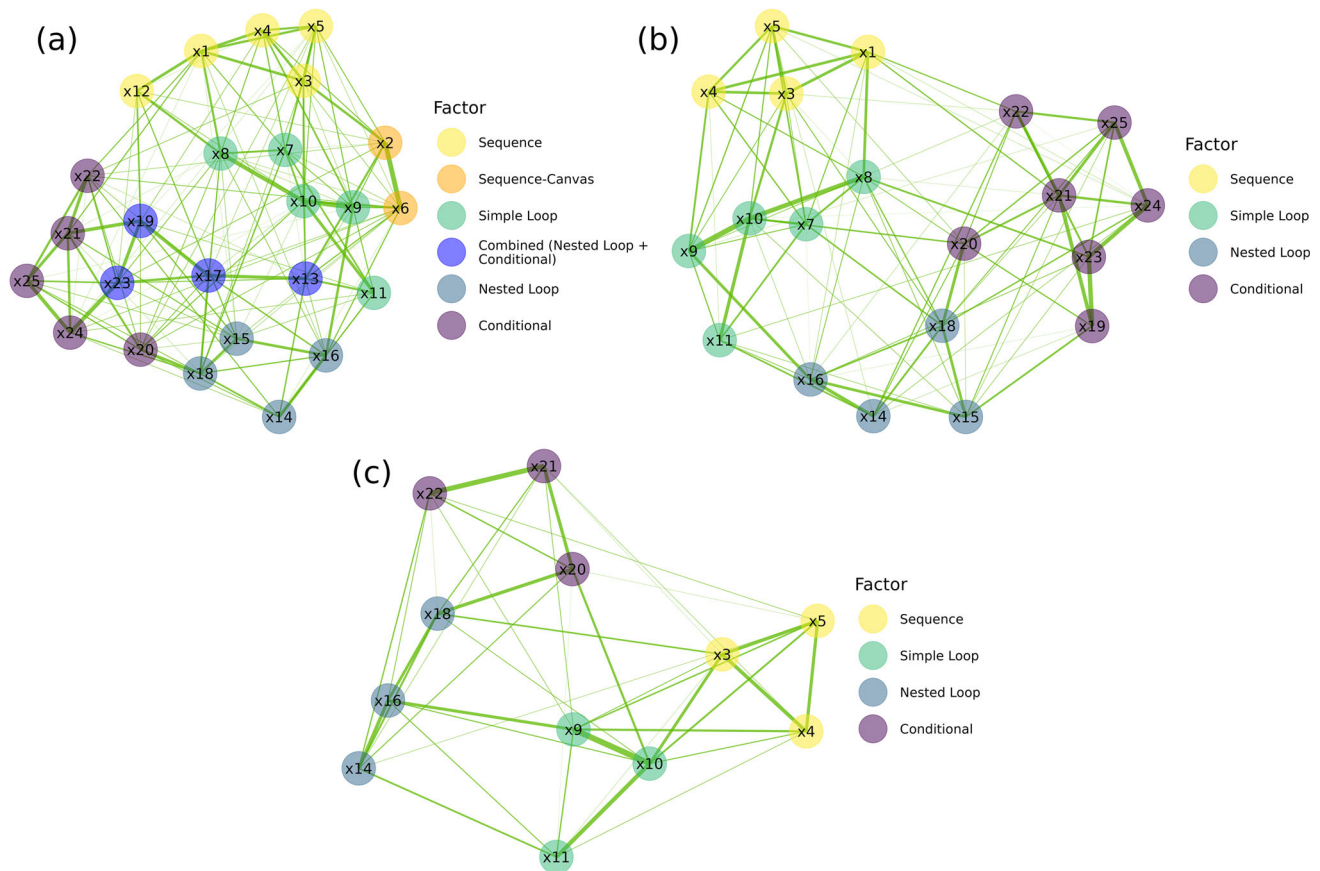
computed the difference in sum scores between the examinees that answered all the items and those who withdrew after answering each item (Diff; Table Table 2). As expected, the highest difference was found after answering item 15, when the examinees that withdrew, on average, had 7.81 less correct answers than those who completed the whole test. In most of the remaining items, the ratio was also positive but smaller.

## Hierarchical Exploratory Graph Analysis

The hierEGA results on the full test found six dimensions that closely resembled the theory upon which the BCTt was developed (Fig. 5a). In addition, the six dimensions were themselves clustered in a single general factor. Only seven problematic items did not fall within their theoretical dimensions. First, the sequence block was split in two clusters, the first containing items 1, 3, 4, 5 and 12 and the second one containing items 2 and 6. Item 12 displayed very weak partial correlations with the remaining items of the nested loop block, whereas items 2 and 6 were canvas-type items and share at least two response alternatives with a similar feature, named taking a wrong direction (error profile 1). These commonalities between items 2 and 6 may have been the reason for their strong association and subsequent grouping into a unique cluster. Second, two items from the nested loop block (i.e., 13 and 17) and two from the conditional block (i.e., 19, 23) merged in a single cluster. Items 13 and 17 were also canvas-type and appeared to be weakly related to the remaining items in the nested loop cluster. An inspection of their response alternatives revealed that they shared the same error profile than the previous canvas-type items (i.e., error profile 1). Items 19 and 23, on the hand, were more similar in that they were built upon a  $3 \times 3$  grid, whereas the other

**Fig. 4** Proportion of correct responses in the BCTt by item and group. The dotted lines indicate the proportions at .90 and .30, respectively, whereas the segments through each data point indicate the limits of the 95% confidence intervals

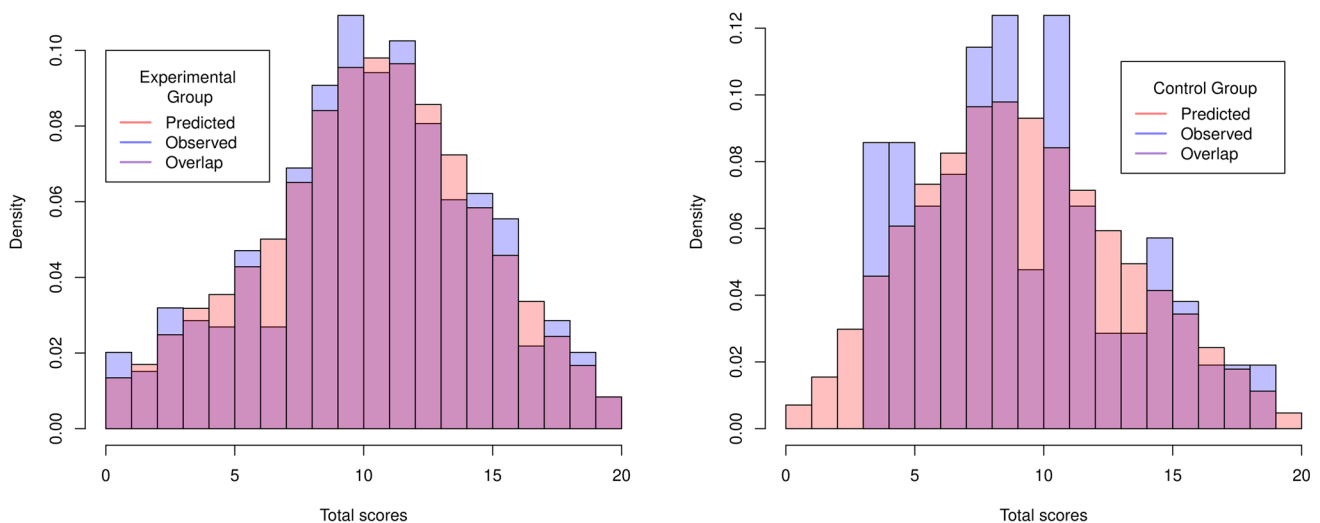




**Fig. 5** Item depuration of the BCTt. (a): Exploratory graph analysis of the 25 items of the BCTt; (b) Exploratory graph analysis of the 20-item selection from the BCTt; (c) Exploratory graph analysis with the final selection of 12 items composing the BCTt-SF

items in the conditional block were in a  $4 \times 4$  grid. However, the relationship between items 13 and 17 with items 19 and 23 was not obvious, and we could not explain why they grouped into the same cluster. Furthermore, items 19

and 23 appeared to be closely related to the remaining items of the conditional cluster to which they theoretically belong. For these reasons, we decided to keep items 19 and 23 in the forthcoming analyses and discarded all the other problematic

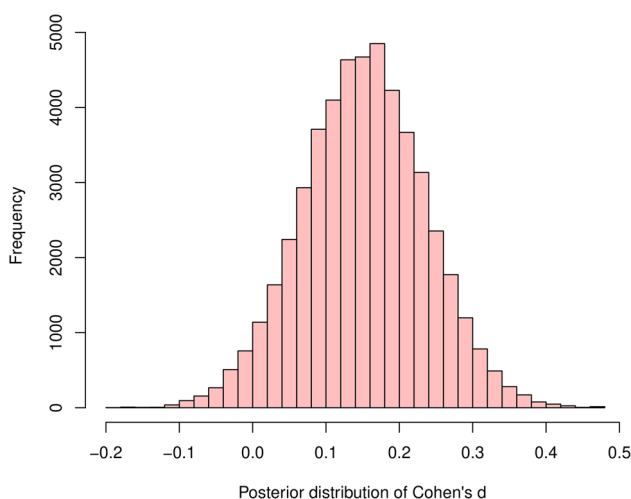


**Fig. 6** Posterior predictive checks for the total scores in the quasi-experimental and control groups

**Table 3** Expected a posteriori estimates and 95% credible intervals (within parentheses) for the bi-factor MIRT model using the 20-item selection

| Item | Factor      | Discrimination     |                    | Location              | Guessing           | $\eta$               |
|------|-------------|--------------------|--------------------|-----------------------|--------------------|----------------------|
|      |             | General factor     | Specific factor    |                       |                    |                      |
| 1    | Sequence    | 1.88 (1.31 - 2.64) | 0.81 (0.17 - 1.56) | 1.96 (1.18 - 2.90)    | 0.31 (0.03 - 0.56) | 0.74 (-0.06 - 1.52)  |
| 3    |             | 1.98 (1.42 - 2.77) | 0.85 (0.18 - 1.63) | 1.70 (1.01 - 2.41)    | 0.21 (0.02 - 0.42) | 0.68 (-0.02 - 1.42)  |
| 4    |             | 1.64 (1.22 - 2.16) | 0.84 (0.23 - 1.57) | 1.62 (1.01 - 2.24)    | 0.10 (0.00 - 0.29) | 0.30 (-0.33 - 0.93)  |
| 5    |             | 1.66 (1.19 - 2.30) | 1.27 (0.43 - 2.31) | 1.28 (0.51 - 1.99)    | 0.09 (0.00 - 0.27) | 0.77 (0.11 - 1.49)   |
| 7    | Simple Loop | 1.93 (1.40 - 2.67) | 0.51 (0.03 - 1.29) | 1.10 (0.43 - 1.71)    | 0.17 (0.01 - 0.36) | 0.72 (0.09 - 1.37)   |
| 8    |             | 2.43 (1.74 - 3.41) | 0.42 (0.02 - 1.14) | 1.11 (0.42 - 1.77)    | 0.18 (0.03 - 0.33) | 0.76 (0.07 - 1.49)   |
| 9    |             | 2.04 (1.54 - 2.70) | 0.59 (0.04 - 1.38) | 1.02 (0.38 - 1.60)    | 0.10 (0.00 - 0.25) | 0.87 (0.25 - 1.53)   |
| 10   |             | 2.59 (1.94 - 3.49) | 0.88 (0.15 - 1.75) | 1.39 (0.75 - 2.15)    | 0.10 (0.00 - 0.26) | 1.22 (0.45 - 2.00)   |
| 11   | Nested Loop | 1.44 (1.00 - 2.09) | 0.71 (0.06 - 1.72) | 1.09 (0.47 - 1.66)    | 0.14 (0.01 - 0.32) | -0.59 (-1.25 - 0.00) |
| 14   |             | 1.34 (0.71 - 2.08) | 1.15 (0.31 - 2.15) | -2.17 (-3.13 - -1.24) | 0.21 (0.07 - 0.31) | 1.03 (0.20 - 1.86)   |
| 15   |             | 1.87 (1.18 - 2.67) | 0.70 (0.10 - 1.42) | -2.32 (-3.34 - -1.49) | 0.10 (0.05 - 0.16) | -0.22 (-1.12 - 0.65) |
| 16   |             | 1.82 (1.15 - 2.65) | 1.28 (0.38 - 2.34) | -1.95 (-2.93 - -0.96) | 0.16 (0.04 - 0.27) | 0.79 (-0.07 - 1.60)  |
| 18   | Conditional | 1.89 (1.18 - 2.79) | 0.94 (0.22 - 1.76) | -2.16 (-3.14 - -1.27) | 0.18 (0.08 - 0.26) | 0.39 (-0.56 - 1.26)  |
| 19   |             | 1.74 (0.83 - 2.71) | 1.82 (0.98 - 2.82) | -1.77 (-2.79 - -0.71) | 0.48 (0.37 - 0.56) | -0.24 (-1.42 - 0.89) |
| 20   |             | 1.97 (1.28 - 2.80) | 0.63 (0.10 - 1.29) | -2.02 (-3.07 - -1.17) | 0.16 (0.09 - 0.23) | -0.32 (-1.22 - 0.56) |
| 21   |             | 1.92 (1.22 - 2.81) | 2.04 (1.33 - 2.91) | -1.78 (-2.71 - -0.93) | 0.09 (0.02 - 0.16) | -0.54 (-1.53 - 0.34) |
| 22   |             | 1.60 (0.93 - 2.37) | 1.15 (0.52 - 1.88) | -1.95 (-2.98 - -1.09) | 0.18 (0.10 - 0.25) | -0.42 (-1.42 - 0.50) |
| 23   |             | 1.96 (1.11 - 2.92) | 1.90 (1.04 - 2.98) | -1.53 (-2.56 - -0.20) | 0.49 (0.35 - 0.58) | 0.11 (-1.08 - 1.20)  |
| 24   |             | 0.79 (0.17 - 1.53) | 2.01 (1.17 - 3.07) | -1.77 (-2.75 - -0.79) | 0.20 (0.09 - 0.28) | -0.75 (-1.89 - 0.24) |
| 25   |             | 1.26 (0.68 - 1.96) | 1.62 (0.98 - 2.40) | -1.96 (-3.06 - -1.03) | 0.19 (0.07 - 0.29) | 0.62 (-0.26 - 1.50)  |

items (i.e., 2, 6, 12, 13, and 17). All the remaining items from the sequence block (i.e., 1, 3, 4, 5, and 7), the simple loop block (i.e., 7, 8, 9, 10, and 11), nested loop block (i.e., 14, 15, 16, and 18), and conditional block (i.e., 20, 21, 22, 24, and 25) were grouped into well-differentiated clusters and thus were also considered in the forthcoming analyses.

**Fig. 7** Posterior distribution of Cohen's  $d$  for the difference between the quasi-experimental and control groups with respect to their mean general factor abilities

A subsequent analysis with hierEGA on the reduced 20-item selection test revealed that the conceptual skills of preschool children could be separated into four well-defined dimensions (i.e., sequences, simple loops, nested loops, and conditionals) and a single general ability indicating a general factor of computational concepts. This time, all the items lied on their expected clusters, confirming that this 20-item selection was accurate enough to ensure a theoretically meaningful structure (Fig. 5b). Moreover, a single general dimension was again suggested in this second dimensionality analysis.

To further reduce the length of the test, we assessed the properties of the items to retain the most informative ones. For this purpose, we used the bi-factor MIRT model.

### Multidimensional Item Response Theory Model

We fitted the bi-factor MIRT model in Eq. 1 using the 20-item selection of the BCTt.<sup>5</sup> The model revealed that the location parameters of most of the items from the sequence, simple loop, and nested loop factors were higher in the quasi-experimental group. In particular, for items 1, 3, 5, 7, 8, 9, 10, 14, and 16, the credible intervals for the effect of the intervention either excluded the zero value or was close to

<sup>5</sup> The diagnostic results for the 20-item selection of the BCTt and the BCTt-SF can be found in Table S1 and Table S2 from the supplemental material, respectively.



excluding it, meaning that it is very probable that these items were easier for the quasi-experimental than for the control group.

The posterior predictive checks indicated a good model fit for both the quasi-experimental and control groups. The predicted probabilities of success of the model matched the observed ones or were close to them (Table S3), and the posterior predictive distribution of the total scores resembled those of the quasi-experimental and control groups (Fig. 6).

The estimated parameters of the bi-factor MIRT model are presented in Table 3. The discrimination parameter estimates presented higher values on the specific factors for more difficult items and higher discrimination values on the general factor for easier items. The location parameter estimates indicated that the items were most discriminant for children in the modest-to-low percentiles of the sequence and simple loop ability levels and in the moderate-to-high percentiles of the nested loop and conditional ability levels.

To compare the quasi-experimental and control groups, we extracted the latent ability estimates of both groups. On average, the quasi-experimental group had higher scores on the general factor than the control group (Fig. 7; Cohen's  $d = 0.15$ ; 95% credible interval =  $-0.02$ – $0.32$ ). Specifically, the probability that the effect was larger than zero was .96. On the other hand, the differences between the specific latent abilities were close to zero.

With respect to the probability of succeeding an item by random chance (i.e., guessing parameters), we found estimates ranging from .09 (items 5 and 21) to .49 (item 23). Interestingly, item 19 also had a similar high guessing parameter estimate (.48). This explains why items 23 and 19 were easy in comparison to the remaining items from the conditional factor. On the one hand, the easiness of item 23 could be explained by response alternatives that were too easy to spot. On the other hand, the correct alternative for item 19 was confounded with a simple sequence. In other words, ignoring the conditional statements and simply following the sequential pattern yielded the correct response. Finally, item 1 also displayed a high guessing probability (.31).

Based on the overall results, we proceeded to select the best three items from each computational factor. From the sequence factor, we selected items 3, 4, and 5, as they presented good discrimination values of the general factor, are informative for a range of specific abilities higher than that of item 1, and did not display high guessing probabilities (guessing  $\leq .21$ ). From the simple loop factor, we decided to retain items 9, 10, and 11 by discarding items 7 and 8, which had the largest guessing probabilities in the simple loop factor (i.e., guessing = .17 and guessing = .18, respectively). From the nested loop block, we selected items 14, 16, and 18. We decided to discard item 15 because it was informative for the general factor only for very high-percentile students. Additionally, the descriptive analyses indicated that

the abrupt difficulty of item 15 could be responsible for test withdrawal. Finally, we selected items 20, 21, and 22 from the conditional factor because they had medium-to-high discrimination values of the general factor and the remaining items either presented high guessing probabilities (i.e., .48 and .49 for items 19 and 23, respectively) or were weaker indicators of the general factor (i.e., discrimination = 0.74 for item 24 and discrimination = 1.26 for item 25).

### Psychometric Properties of the Short-Form of the Beggins Computational Thinking Test

To confirm the adequate psychometric properties of this final 12-item selection, we repeated the dimensionality assessment with hierEGA (Fig. 5c) and estimated the bi-factor MIRT model in Eq. 1 (Table 4). As expected, hierEGA yielded a graph whose clusters corresponded to the four theoretical computational concepts and a single general factor of computational concepts. The estimated parameters from the bi-factor MIRT model were similar to the ones estimated in the 20-item version. No item had a high guessing probability. Again, with high probability, items 3, 5, 9, 10, 14, and 16 were easier for the quasi-experimental group. The item characteristic and information curves for the BCTt-SF for both the quasi-experimental and control groups are presented in Figures S2–S5 from the Supplemental material.

Finally, we used the calibration of the bi-factor MIRT model to create an online application in the Shiny platform that can be used to automatically score examinees. The app can be accessed at <https://computationalthinkingtests.shinyapps.io/SF-BCTt/>. The user only needs to upload a file in *csv* format with a matrix of dichotomous responses (i.e., 1 for a correct response and 0 for an incorrect response), where the participants are in rows and the items are in columns.

## Discussion

CT is now widely recognized as a fundamental subject that should be incorporated into compulsory education at all grades (Georgiou & Angeli, 2019; Stamatios, 2022). An important part of the process of teaching CT is the explanation and application of computational concepts. These concepts extend beyond the realm of technology and are relevant in various aspects of everyday life. In this context, sequences, loops, and conditionals are three of the most popular computational concepts in teaching programs for preschool students (McCormick & Hall, 2022). However, the instruments that are currently used to assess these concepts are usually created ad-hoc and did not undergo a proper psychometric validation (Tang et al., 2020). To overcome this lack of reliable CT instruments in preschool research, we developed a short-form version of the BCTt that displayed



**Table 4** Expected a posteriori estimates and 95% credible intervals (within parentheses) for the bi-factor MIRT model of the BCTt-SF

| Item | Factor      | Discrimination     |                    | Location              | Guessing           | $\eta$               |
|------|-------------|--------------------|--------------------|-----------------------|--------------------|----------------------|
|      |             | General factor     | Specific factor    |                       |                    |                      |
| 3    | Sequence    | 1.90 (1.40 - 2.63) | 0.87 (0.16 - 1.61) | 1.48 (0.75 - 2.27)    | 0.23 (0.02 - 0.45) | 0.82 (0.08 - 1.56)   |
| 4    |             | 1.74 (1.30 - 2.30) | 0.85 (0.17 - 1.58) | 1.42 (0.75 - 2.10)    | 0.15 (0.01 - 0.36) | 0.44 (-0.23 - 1.11)  |
| 5    |             | 1.71 (1.24 - 2.29) | 1.11 (0.25 - 2.10) | 1.08 (0.22 - 1.75)    | 0.12 (0.01 - 0.32) | 0.90 (0.24 - 1.64)   |
| 9    | Simple Loop | 2.07 (1.60 - 2.76) | 0.55 (0.04 - 1.30) | 1.11 (0.48 - 1.70)    | 0.07 (0.00 - 0.20) | 0.89 (0.22 - 1.57)   |
| 10   |             | 2.21 (1.65 - 3.09) | 0.63 (0.06 - 1.40) | 1.30 (0.68 - 1.98)    | 0.08 (0.00 - 0.24) | 1.11 (0.41 - 1.84)   |
| 11   |             | 1.61 (1.15 - 2.25) | 0.65 (0.04 - 1.51) | 1.16 (0.50 - 1.74)    | 0.13 (0.01 - 0.30) | -0.56 (-1.20 - 0.06) |
| 14   | Nested Loop | 1.54 (0.83 - 2.18) | 1.26 (0.39 - 2.16) | -2.15 (-3.14 - -1.21) | 0.21 (0.08 - 0.32) | 0.94 (0.10 - 1.81)   |
| 16   |             | 1.82 (1.22 - 2.53) | 1.26 (0.41 - 2.16) | -1.92 (-2.86 - -0.95) | 0.16 (0.04 - 0.26) | 0.84 (-0.01 - 1.63)  |
| 18   |             | 1.76 (1.08 - 2.64) | 1.02 (0.24 - 1.74) | -2.20 (-3.23 - -1.27) | 0.19 (0.09 - 0.27) | 0.40 (-0.49 - 1.30)  |
| 20   | Conditional | 1.86 (1.24 - 2.62) | 1.06 (0.25 - 1.84) | -2.11 (-3.18 - -1.20) | 0.17 (0.09 - 0.23) | -0.31 (-1.25 - 0.61) |
| 21   |             | 1.79 (1.12 - 2.56) | 1.49 (0.46 - 2.78) | -1.90 (-2.86 - -0.98) | 0.14 (0.04 - 0.22) | -0.49 (-1.50 - 0.40) |
| 22   |             | 1.67 (0.99 - 2.35) | 1.27 (0.37 - 2.19) | -2.03 (-3.05 - -1.12) | 0.17 (0.09 - 0.25) | -0.28 (-1.25 - 0.63) |

excellent psychometric qualities. More concretely, we successfully performed item revision to select the best three items from each computational dimension assessed while maintaining optimal reliability properties, yielding a short-form version with 12 items (BCTt-SF).

During the data analysis process, we were able to elucidate questions regarding the computational concepts that can be assessed at a preschool age. We found that the four computational concepts measured by the BCTt (i.e., sequences, simple loops, nested loops, and conditionals) were differentially defined, confirming that preschoolers can be differentiated according to these four concepts (e.g., some of them understand the concepts and others do not).

We replicated previous research regarding the good performance of preschoolers when dealing with sequences and interpreting simple loops (Angeli & Valanides, 2020; Relkin & Bers, 2021; Saxena et al., 2020), but, more interestingly, we found that nested loops and conditionals were much more difficult and only some preschoolers could understand them. This finding is consistent with previous research showing that preschoolers' ability to display accurate conditional reasoning can vary depending on the specific context of the task (Andrews et al., 2012; Diamond et al., 2005).

In addition to the four computational concepts identified, a general factor reflecting an overall ability of computational concepts was supported. This result was highly desirable as it justified the computation of a single test score, which is much more reliable than the scores of each specific computational concept. We also found that the items from the sequence and simple loop blocks were more related to the general factors than those from the nested loop and conditional blocks. That is, the general factor seemed to be more involved in the solution of easier rather than more difficult items. This result

was sensible because nested loops and conditionals are much more nuanced concepts and hence that may require specific skills to understand them correctly.

We also tested the effectiveness of an educational intervention based on unplugged methodologies. Unplugged approaches to teaching CT are crucial in preschool settings because the plugged ones could be too difficult for them: some children may not have enough prior exposure to technological devices and may not adapt well to coding activities within a computational environment. Additionally, economic constraints may limit their access to such technologies. Given these challenges, unplugged methods provide an excellent alternative. Specifically, we determined which computational concepts are more likely to be fostered after an unplugged educational intervention. We found that the quasi-experimental group was consistently superior to the control group in the understanding of sequences, simple, and nested loops. However, the educational intervention seemed to be more beneficial for improving the understanding of easier rather than difficult concepts. The highest improvements were observed for sequences and simple loops, whereas the improvement in the nested loop block was observed to a lesser extent and no improvement was observed for the conditional block. This result suggested that understanding conditional statements may require more specific training.

Some limitations are worth noting. First, sample demographics are lacking, so differential item functioning depending on characteristics like sex could not be assessed. Notwithstanding, the fact that the bi-factor model already presented excellent fit indices and the organization of the data largely matched the theoretical dimensionality of the test are signs that these characteristics would not play a major explanatory role in the model.

Second, the pattern of missing responses was mostly driven by test withdrawal probably due to both fatigue and discouragement of low-performing examinees. This could introduce some misspecification in the statistical model that we fitted. A possible solution to this could be to generalize the bi-factor model to explicitly model the probability of withdrawal and relate it to the ability of the examinees. Another problem of progressive test withdrawal is that it could prompt systematic correlations between items that are proximal in the test, a phenomenon that was also ignored. Unfortunately, the current study neither provides information on how the results can be linked to specific teaching methods and the teachers' characteristics.

Third, we did not elucidate whether the general factor of computational concepts assessed by the BCTt-SF actually corresponds to a measure of general intelligence (i.e., discriminant validity). Previous research showed that CT skills are correlated with measures of intelligence like spatial, reasoning, and problem-solving abilities (Román-González et al., 2017, 2018) and also relate to students' numerical and verbal abilities (Tsarava et al., 2022).

A final limitation of this work is that the BCTt-SF is focused on computational concepts and practices and the assessment of computational perspectives is lacking (Brennan & Resnick, 2012). Computational perspectives are abilities that allow children to share information and express ideas, allowing them to teach others, interact, collaborate, discuss, and bring into question the surrounding technological world. In other words, these abilities are the most socially oriented as they bear the purpose of entertaining, educating, engaging with others, and critically question what and how products can be improved. Eventually, the achievement of these goals will yield computational fluency, which is the capacity to understand computational concepts and applying their meaning to express oneself creatively and effortlessly with digital technologies (Stamatios, 2021). Therefore, additional methods for assessing the progress of computational perspectives and computational fluency are required to get a broader picture of CT skills (Román-González, 2019).

Despite these limitations, the current study has several strengths: we used a sample of 700 children, the largest up to date in any CT research with preschool children. This allowed us to study the competence of children in the most common computational concepts assessed in the literature, named sequences, loops, and conditionals. We replicated previous results that reported that children can solve with high accuracy problems involving simple and complex sequences and we also found that they performed well with simple loops, but presented more difficulties to understand nested loops and conditionals. We also carried out several psychometric analyses to demonstrate the internal validity and adequate reliability of both the 20-item version of the BCTt and the 12-item version composing the BCTt-SF. Additionally, it is

crucial to highlight that the educational intervention implemented in our study was not tailored to specifically target the content covered in the BCTt. Therefore, the results obtained from the BCTt should reflect the generalizable properties of CT learning rather than being influenced by specific knowledge gained through test training.

The BCTt-SF can serve as a valuable tool to assess the effectiveness of new educational interventions and to compare them. It can provide insights into the efficacy of different approaches in fostering computational concepts. It is important to note that the BCTt-SF can be used to test the efficacy of new educational interventions or compare them in an objective way. It should be noted that the unplugged activities carried out during the educational intervention were not specifically designed to target the content covered in the BCTt (e.g., identifying patterns in a grid). In other words, the higher scores observed in the quasi-experimental group cannot be attributed to the similarity between the unplugged activities of the intervention and the BCTt items.

Empirical studies concerning preschool students often use ad-hoc methods for assessing the computational thinking skills of the children (Bakala et al., 2021; McCormick & Hall, 2022; Tang et al., 2020). Given the lack of well-validated instruments for measuring these abilities in preschool students, the BCTt-SF sets an unprecedented benchmark towards the rigorous scoring of preschool students and for testing and comparing educational interventions in their understanding of sequences, loops, and conditionals (El-Hamamsy et al., 2022a; Vourletsis & Politis, 2024). Whereas multiple tests exist for measuring CT skills in lower-primary school (Li et al., 2021; Relkin et al., 2020; Zapata-Cáceres et al., 2021; Zhang & Wong, 2023), they are lacking in preeschool settings and do not cover the aforementioned computational concepts (Relkin & Bers, 2021). Researchers and practitioners will benefit from this new assessment instrument for several reasons: (1) the BCTt-SF is a short yet reliable instrument; (2) the relevance and boundaries of the measured dimensions are clearly established; (3) the scoring system was designed according to the validated underlying statistical model and can be automated via an online application so that researchers and practitioners can seamlessly score their examinees; (4) the test is unplugged and independent from the activities and games commonly used in educational interventions like the Tangible Robotics Program (Bers et al., 2014), guarding against biases due to familiarity, practice, and coding knowledge.

In future research, it would be valuable to explore additional forms of validity for the BCTt-SF, such as convergent and discriminant validity in relation to other computational thinking (CT) concepts. A potential approach could involve utilizing both the BCTt-SF and the TechCheck-K assessments in conjunction to gain a comprehensive understanding of students' CT competence. This would enable the examina-

tion of the interrelationships between all the computational concepts being measured. Another aspect worth investigating is the predictive validity of the BCTt-SF. For instance, gathering data on students' performance in standard curriculum subjects and correlating them with their CT scores could provide insights into how CT relates to academic achievement.

Finally, for researchers and practitioners planning to use the BCTt-SF in a preschool setting, it is important to administer the test in a physical format (i.e., paper) and to precede the test with examples. For each computational concept, the evaluator should read a representative item aloud, explain how to choose an option, and demonstrate the reasoning behind the correct choice. We also recommend allowing ample time for the examinees to read and answer all the test items. Preschoolers may sometimes lose patience or motivation or may simply withdraw from the test due to fatigue. Therefore, it is crucial to encourage examinees to answer all the items, even if they are unsure of the correct answer. This approach is recommended because the statistical model that we used does not accommodate missing responses, so all answers must be coded as correct or incorrect. This is not problematic, as the model accounts for the likelihood of correct answers occurring by chance.

## Conclusion

An important component of the CT learning curricula is the adequate monitoring of the student progress, for which standardized, valid, and reliable psychometric instruments are required. Unfortunately, instruments with such properties for the assessment of CT skills are lacking in preschool settings. For the first time, we explicitly addressed this issue by developing a 12-item test (BCTt-SF) in a sample of 700 preschool students, obtaining excellent internal validity and reliability properties. Using the best items from the BCTt, we confirmed four well-defined, nonoverlapping factors to be independently assessed in preschoolers (i.e., sequences, simple loops, nested loops, and conditionals) and a single general factor that may reflect an overall competence in the understanding of these computational concepts. From these factors, we found that an educational intervention based on unplugged methodologies may improve the understanding of sequences and simple loops, whereas improving the understanding of nested loops and conditional statements may require more specific training at this early age. Finally, to ensure the achievement of reliable scores for preschool students, the BCTt-SF test was made freely available and can be automatically scored via an online application at <https://computationalthinkingtests.shinyapps.io/SF-BCTt/>.

## Appendix

### Confirmatory Factor Analyses

We sought to identify the most reliable and informative items. For this purpose, we used the graph obtained with hierEGA to define a model to be fitted with Confirmatory Factor Analysis (CFA).<sup>6</sup> Factor loadings with an absolute value lower than .30 were considered small, those between .30 and .50 were considered medium, and larger values were deemed to be high. As the observed scores were dichotomous (1: correct response; 0: wrong response), we used the tetrachoric correlation matrix (i.e., the correlation matrix between the latent continuous scores) to fit the CFA models.<sup>7</sup>

Two types of models were estimated with CFA: a correlated factor model, in which the first-order clusters (i.e., specific factors) detected by hierEGA were allowed to correlate, and a bi-factor model, a popular method to address within-item multidimensionality by decomposing systematic item variance in a general factor that explains the commonalities between all the items and specific factors that only affect a subset of the items (Reise, 2012). If a general factor existed according to hierEGA, then we would expect medium or high correlations among all the specific factor correlations from the correlated factors model. If so, the bi-factor model could be used to quantify how much information each item contributes to measure such a general factor. Finding a general factor is very desirable because it would mean that computing a single test score reflecting computational concepts knowledge is possible. Furthermore, the reliability of such a score would be much higher than any of the specific abilities because all the items would be expected to load on this general factor.

In the bi-factor model, we assessed the reliability of the general factor score with omega hierarchical ( $\omega_H$ ; Garcia-Garzon et al., 2019), a statistic indicating the ratio of the variance accounted for by the general factor to the test variance. According to Rodriguez et al. (2015), it is considered

<sup>6</sup> Note that hierEGA is an exploratory tool concerned with the estimation of the number of factors in the data, whereas CFA is confirmatory and provides the measurement model with the regression weights of the factors on the items. As such, they do not provide redundant information and have been recommended to be used in tandem (Golino & Demetriou, 2017a).

<sup>7</sup> We also used exploratory factor analysis (EFA) with the unweighted least squares estimator and target rotation to check the agreement of the loading pattern with the clustering structure of hierEGA. However, many negative cross-loadings were estimated across the structure, wrongly implying that a higher ability level in some traits is detrimental to others. Therefore, we decided to interpret only the results of the CFA models.

that values over .80 indicate that the general factor is a very reliable source to create a single score, as most of the test variance is attributed to it.

We used three fit indices to assess the adequacy of the CFA models: The unbiased root mean square residual (uSRMR; Maydeu-Olivares, 2017), the root mean square error of approximation (RMSEA; Steiger, 2000), and the Comparative Fit Index (CFI; Bentler, 1990). The uSRMR measures how close the CFA model predicts the observed correlation matrix between the items, whereas the RMSEA estimates the distance between the expected model correlation predictions and the true ones in the population. Lastly, the CFI is a measure comparing the fit of the estimated model with that of a null model where no latent factors are present. Values of uSRMR and RMSEA between .05 and .08 are usually regarded as indicating an acceptable fit and values lower than .05 indicate a good fit (Maydeu-Olivares, 2017; McDonald & Ho, 2002). On the other hand, the cutoff values for the CFI are .90 or greater for an acceptable fit and .95 or greater for a good fit (Hu & Bentler, 1999).

An important caveat regarding these fit indices is that the cutoffs used to assess model's goodness-of-fit are somewhat arbitrary because they were established in specific contexts (e.g., for specific statistical models, estimators, number of items) and may not be adequate in many applied situations. Thereby, we also assessed the adequacy of the CFA mod-

els by inspecting the modification indices (MI) to detect possible sources of residual dependencies. MI are  $\chi^2$  statistics that indicate the amount of model fit improvement if a parameter that was not estimated was actually estimated in turn. These statistics may identify different types of model misspecification like residual dependencies between some items, items loading on multiple factors, and correlations between originally assumed orthogonal factors. A significant fit improvement is attained when the MI > 3.84, value that corresponds to the 95th percentile of the distribution of a  $\chi^2$  statistic with one degree of freedom. However, higher values are usually expected because of either sampling error (i.e., the number of false positives may be high due to the many parameters being tested) or external factors not considered in the model. Therefore, we only interpreted the MI that were significant at the 99% confidence level (MI > 6.63).

Overall, the CFM presented excellent absolute and relative fit indices (uSRMR[CFM]: .040 (95%CI: .031-.049); RMSEA[CFM]: .019 (95%CI: .006-.028); CFI[CFM]: .994). The CFM revealed medium-to-high factor loadings on all the items (Table 5), ranging from .54 (i.e., item 24 on the conditional factor) to .87 (i.e., item 10 on the simple loop factor). The factor correlation estimates (Fcor; Table 6) indicated that the sequence factor was strongly correlated with the ability to solve simple loops problems (Fcor = .84), which was related to the nested loop factor (Fcor = .73). The latter was also correlated with the conditional factor (Fcor = .63), creating an

**Table 5** Factor loading estimates and 95% confidence intervals (within parentheses) for the correlated factors and bi-factor models using the 20-item selection

| Item | Factor      | Correlated model   | Bi-factor model        |                         |
|------|-------------|--------------------|------------------------|-------------------------|
|      |             | Factor loading     | General factor loading | Specific factor loading |
| 1    | Sequence    | 0.72 (0.61 – 0.82) | 0.56 (0.45 – 0.67)     | 0.49 (0.31 – 0.67)      |
| 3    |             | 0.78 (0.69 – 0.86) | 0.62 (0.53 – 0.71)     | 0.44 (0.28 – 0.61)      |
| 4    |             | 0.73 (0.65 – 0.82) | 0.57 (0.48 – 0.67)     | 0.46 (0.30 – 0.63)      |
| 5    |             | 0.72 (0.64 – 0.81) | 0.57 (0.47 – 0.66)     | 0.44 (0.27 – 0.60)      |
| 7    | Simple loop | 0.73 (0.66 – 0.81) | 0.72 (0.64 – 0.80)     | 0.08 (–0.14 – 0.30)     |
| 8    |             | 0.81 (0.74 – 0.87) | 0.82 (0.74 – 0.89)     | 0.01 (–0.21 – 0.24)     |
| 9    |             | 0.79 (0.72 – 0.86) | 0.75 (0.67 – 0.83)     | 0.27 (0.06 – 0.49)      |
| 10   |             | 0.87 (0.82 – 0.93) | 0.83 (0.75 – 0.91)     | 0.43 (0.13 – 0.72)      |
| 11   | Nested loop | 0.61 (0.53 – 0.69) | 0.57 (0.48 – 0.66)     | 0.28 (0.04 – 0.51)      |
| 14   |             | 0.55 (0.45 – 0.65) | 0.42 (0.32 – 0.53)     | 0.40 (0.21 – 0.59)      |
| 15   |             | 0.61 (0.50 – 0.72) | 0.48 (0.36 – 0.60)     | 0.35 (0.16 – 0.54)      |
| 16   |             | 0.73 (0.65 – 0.81) | 0.58 (0.49 – 0.67)     | 0.45 (0.25 – 0.64)      |
| 18   | Conditional | 0.68 (0.59 – 0.77) | 0.55 (0.46 – 0.64)     | 0.37 (0.19 – 0.54)      |
| 19   |             | 0.61 (0.53 – 0.70) | 0.35 (0.25 – 0.45)     | 0.50 (0.40 – 0.61)      |
| 20   |             | 0.66 (0.57 – 0.76) | 0.51 (0.41 – 0.62)     | 0.28 (0.16 – 0.41)      |
| 21   |             | 0.76 (0.68 – 0.84) | 0.45 (0.34 – 0.55)     | 0.62 (0.52 – 0.72)      |
| 22   |             | 0.63 (0.53 – 0.72) | 0.39 (0.28 – 0.50)     | 0.46 (0.34 – 0.57)      |
| 23   |             | 0.68 (0.60 – 0.75) | 0.37 (0.28 – 0.47)     | 0.61 (0.51 – 0.71)      |
| 24   |             | 0.54 (0.45 – 0.63) | 0.23 (0.11 – 0.34)     | 0.61 (0.50 – 0.73)      |
| 25   |             | 0.65 (0.56 – 0.73) | 0.34 (0.24 – 0.44)     | 0.60 (0.50 – 0.70)      |

**Table 6** Factor correlation estimates in the correlated factors model using the 20-item selection

| Factor      | Sequence | Simple loop | Nested loop | Conditional |
|-------------|----------|-------------|-------------|-------------|
| Sequence    | —        | —           | —           | —           |
| Simple loop | .84      | —           | —           | —           |
| Nested loop | .48      | .73         | —           | —           |
| Conditional | .39      | .52         | .63         | —           |

interdependence among the factors that could be explained by the presence of a single general factor. In fact, the smallest correlation, between the sequence and conditional factors, was medium-sized ( $F_{cor} = .39$ ).

This result, coupled with the results obtained from the hierEGA, prompted us to estimate item loadings on a general factor by means of the BFM. The BFM also presented excellent fit values (uSRMR[CFM]: .047 (95% CI: .038–.057); RMSEA[CFM]: .026 (95% CI: .017–.035); CFI[CFA]: .990), complying with the standards of goodness-of-fit. The loadings on the general factor ranged from small to high (Table 5), the highest being in the simple loop factor (mean = .74), followed by the sequence factor (mean = .58), nested loop factor (mean = .51), and conditional factor (mean = .38). In other words, the pattern of loadings resembled those of the discrimination parameter estimates in the bi-factor MIRT model, with higher loadings on the specific factors for more difficult items and higher loadings on the general factor for easier items.

The reliability of the general factor was adequate ( $\omega_H = .79$ ), indicating that nearly 80% of the test variance could be summarized in a single general factor score. On the other hand, the loadings on the specific factors were small-to-medium, presenting higher values in the items from the conditional block, and higher in the nested loop block than in the simple loop block.

The BFM revealed some candidate items for removal. The weakest-defined specific factor was the simple loop factor, where items 7 and 8 had loadings close to zero (i.e., .08 and .01, respectively) and items 9 and 11 displayed low loadings (i.e., .27 and .28, respectively). Item 20 also loaded weakly on the conditional factor (i.e., .28). All the remaining items presented medium or high loadings on their respective specific factors. With regards to the general factor, all the items in the sequence, simple loop, and nested loop blocks had either medium or high loadings on the general factor, whereas most items from the conditional block displayed medium loadings. Only item 24 had a low loading on the general factor (i.e., .23).

Next, we looked at the MI of the BFM to identify possible sources of model misspecification. The MI indicated that the BFM could be significantly improved if the nested loop factor was allowed to correlate with the conditional factor (MI = 52.21). Similarly, the sequence factor could be correlated

with both the loop nested factor (MI = 29.20) and the simple loop factor (MI = 20.70), whereas the conditional factor could be correlated with both the simple loop factor (MI = 9.10) and the sequence factor (MI = 7.64). The only factor correlation that could not significantly improve the fit was that between the simple loop and nested loop factors. As it was shown before, withdrawing the test became more common from the nested loop block onward, so residual dependencies between the nested loop and conditional factors could be explained by the presence of these many absent responses. On the other hand, if the correlation between the sequence and nested loop factors was estimated, then it would be negative, which is compatible with the fact that some low-performing children withdrew when arriving at the nested loop block.

With regards to cross-loadings, the MIs indicated that estimating many of them could also significantly improve the fit of the bi-factor model. However, we could not figure out an explanation for them (e.g., allowing item 14 to load on the conditional factor would lead to a fit improvement of MI = 18.41, but item 14 does not contain any conditional instruction). Finally, we found residual correlations between items 20 and 18 (MI = 11.92), 19 and 15 (MI = 9.57), 20 and 5 (MI = 7.30), and 23 and 19 (MI = 6.86). After inspecting all these item pairs in search for similarities, we did not find any apparent relationship in terms of content or difficulty except for items 23 and 19. Both of these were much more easier than the remaining items from the conditional factor, possibly because the response alternatives were bad and children could spot the correct answer by guessing.

To confirm the adequate psychometric properties of this final 12-item selection, we refitted the BFM (Table 7). The BFM displayed again excellent absolute and relative fit indices (uSRMR: .51 (95%CI: .38–.65); RMSEA: .037 (95% CI: .022–.051, CFI: 0.986)) and the reliability of the general factor was also good ( $\omega_H = .80$ ). Regarding the loadings, only items 9 and 11 had low loadings on their specific factor (i.e., values of .11 and .14 on the simple loop factor, respectively), whereas the remaining item loadings on the general and specific factors were medium or high.

### Bi-Factor MIRT Model

An important detail is that missing responses should not have a parameter accounting for guessing because in these cases,



**Table 7** Factor loading estimates and 95% confidence intervals (within parentheses) for the bi-factor model of the final 12-item selection (BCTt-SF)

| Item | Factor      | General factor loading | Specific factor loading |
|------|-------------|------------------------|-------------------------|
| 3    | Sequence    | 0.62 (0.52, 0.72)      | 0.45 (0.25, 0.65)       |
| 4    |             | 0.57 (0.47, 0.67)      | 0.46 (0.26, 0.66)       |
| 5    |             | 0.56 (0.46, 0.66)      | 0.45 (0.25, 0.65)       |
| 9    | Simple loop | 0.80 (0.71, 0.89)      | 0.11 (−0.33, 0.55)      |
| 10   |             | 0.85 (0.76, 0.94)      | 0.52 (−1.14, 2.17)      |
| 11   |             | 0.61 (0.50, 0.71)      | 0.14 (−0.41, 0.68)      |
| 14   | Nested loop | 0.42 (0.31, 0.52)      | 0.50 (0.25, 0.76)       |
| 16   |             | 0.58 (0.48, 0.67)      | 0.42 (0.20, 0.64)       |
| 18   |             | 0.52 (0.42, 0.63)      | 0.34 (0.14, 0.53)       |
| 20   | Conditional | 0.50 (0.39, 0.61)      | 0.35 (0.18, 0.51)       |
| 21   |             | 0.42 (0.31, 0.53)      | 0.69 (0.43, 0.95)       |
| 22   |             | 0.38 (0.26, 0.50)      | 0.49 (0.28, 0.70)       |

the items were never tried, rendering the probability of succeeding the item to be zero. Furthermore, as many of the missing responses happened because the examinees withdrew the test, we impeded missing values to contribute to the likelihood in the bayesian IRT model. Let  $\Theta$  be a parameter-vector collecting all the parameters of the model (1), so the (Bernoulli) probability mass function of a single observation can be written as

$$P(Y = y|\Theta) = \begin{cases} p, & \text{if } y = 1; \\ 1 - p, & \text{if } y = 0; \\ 0, & \text{if } y \text{ is missing.} \end{cases}$$

The posterior distribution of the parameters is then proportional to

$$P(\Theta|y) \propto P(y|\Theta)P(\Theta),$$

where  $P(\Theta)$  is the prior probability of the parameters, a quantity reflecting our epistemic knowledge before seeing the data.

We defined priors on all the item discrimination parameters (i.e.,  $\alpha_j$  and  $\alpha_{gj}$ ) using a half-normal distribution with a location hyperparameter for each factor and one scale hyperparameter. Following the advice of Bürkner (2021) of using a distribution with mode at zero and then decreasing the probability density for larger values. The discrimination hyperparameters had another half-normal distribution centered at zero and with standard deviation one. We also chose the same half-normal prior for the standard deviation of the location parameters ( $d_j$ ) but a standard normal prior over the means for each factor. To resemble our uncertainty about the guessing parameters ( $c_j$ ), we chose a beta distribution with hyperparameters  $a = 1$  and  $b = 3$ , meaning that from 4 trials, we expected one success and three failures if the examinee answers the item without knowing the correct alternative.

With this distribution, we a priori expected that all the alternatives were equally attractive. Finally, we assigned a standard normal prior distribution to the new  $\eta_j$  parameters.

Finally, we fixed the metric of the latent factors with a normal prior distribution with location zero and standard deviation one. All the factors were assumed to be independently distributed. Overall, the full model configuration was

$$\begin{aligned} (\mu_{\alpha_g}, \mu_{\alpha_s}, \sigma_{\alpha}, \sigma_d) &\sim \text{Normal}(0, 1), (\mu_{\alpha_g}, \mu_{\alpha_s}, \sigma_{\alpha}, \sigma_d) \geq 0, \\ \mu_{d_s} &\sim \text{Normal}(0, 1), \\ \alpha_j &\sim \text{Normal}(\mu_{\alpha_s}, \sigma_{\alpha}), \alpha_j \geq 0, \\ \alpha_{gj} &\sim \text{Normal}(\mu_{\alpha_g}, \sigma_{\alpha}), \alpha_{gj} \geq 0, \\ d_j &\sim \text{Normal}(\mu_{d_s}, \sigma_d), \\ \eta_j &\sim \text{Normal}(0, 1), \\ c_j &\sim \text{Beta}(1, 3), \\ (\theta_g, \theta_s) &\sim \text{Normal}(0, 1). \end{aligned}$$

For estimating the bayesian model (1), we run five chains with a Hamiltonian Monte Carlo algorithm and extracted 22,000 samples from the posterior distribution in each chain, discarding the first 2000 iterations as warmup. In total, 100,000 samples were extracted from the joint posterior distribution. The model setup, posterior convergence, and results were validated following the statistical workflow in Gelman et al. (2020). For instance, we conducted predictive prior checks to ensure that the prior distributions produced meaningful data. More concretely, we generated a prior predictive distribution of the total score. To account for the missing responses, we inserted zeros in the same positions of the simulated datasets than in the real dataset. Figure S1 illustrates this prior predictive check. The mean of the distribution was 10.80, with a standard deviation of 3.58 and interquartile range of [8.70–13.27]. We deemed these statistics to be a reasonable representation of realistic data.

We also inspected the sampling diagnostics of the posterior distributions of the parameters. For assessing the

convergence of the samples to the intended posterior distribution, we computed the *Rhat* statistic, the bulk effective sample size ( $ESS_{bulk}$ ), and the tail effective sample size ( $ESS_{tail}$ ). *Rhat* is a measure of how well the Markov chains mixed. Typically, *Rhat* values less than 1.05 are considered to be good indicators that the Markov chains converged to the same posterior distribution. On the other hand,  $ESS_{bulk}$  is a measure of sampling efficiency for the EAP estimate of the posterior distribution, whereas  $ESS_{tail}$  is the minimum effective sample size for 5% and 95% of the posterior quantiles. Both  $ESS_{bulk}$  and  $ESS_{tail}$  should be at least 100 per Markov chain to have reliable estimates of the posterior quantiles.

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**Data Availability** The dataset used in this study and all the files necessary to reproduce the analyses are available at [https://osf.io/3xn7a/?view\\_only=7190e194469f44119aa9174b768a406d](https://osf.io/3xn7a/?view_only=7190e194469f44119aa9174b768a406d).

## Declarations

**Conflict of Interest** The authors declare no competing interests.

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